

Microprocessor and Embedded System Project Report

School of Applied Science and Technology
Shahjalal University of Science and
Technology

Course code: EEE-704

Project: IoT Based Data Logger

Submitted By

Md. Al Amin Chy (2019338509)

Syed Nadim Mehdi (2019338563)

BSc in Electrical and Electronic Engineering

4th year 1st semester

Session: 2019-2020

Sylhet Engineering College

Submitted To

Md. Janibul Alam Soeb

Assistant Professor

Faculty of Agricultural Engineering and Technology

Sylhet Agricultural University

Sylhet -3100, Bangladesh

Project title: To build an IoT based data logger for measuring environment parameters

Required Component:

- ESP32
- MQ-135
- DHT11
- TEMT6000
- BMP180
- Vero Board and jumper cables
- Power Supply

Working Principle:

The IoT-based data logger employs an ESP32 Dev-Kit microcontroller interfaced with multiple sensors for real-time environmental data acquisition and transmission. The system components include the MQ-135 gas sensor, DHT11 temperature and humidity sensor, TEMT6000 light sensor, BMP180 pressure sensor, a Vero board with jumper cables for connectivity, and a stable power supply.

Sensor data is collected periodically, with the MQ-135 monitoring air quality, the DHT11 measuring temperature and humidity, the TEMT6000 recording ambient light intensity, and the BMP180 tracking atmospheric pressure and altitude. These sensors generate analog or digital outputs, which are processed by the ESP32.

The ESP32's onboard Wi-Fi module enables seamless data transmission. A pre-configured cloud-based script (Google Apps Script) facilitates data upload to Google Sheets every 5 seconds, ensuring continuous and synchronized logging. The Vero board organizes connections, enhancing system reliability.

This implementation offers a robust, cost-effective solution for environmental monitoring with real-time data storage accessible remotely for analysis.

Circuit Diagram:

Firmware:

```
#include <WiFi.h>
#include <HTTPClient.h>
#include "DHT.h"

// Wi-Fi credentials
const char* ssid = "ROOM302";
const char* password = "ROOM_605";

// Google Apps Script Web App URL
const char* serverName = "https://script.google.com/macros/s/AKfycbztrrsGnEYjJ3V0-
pjrA31lWgU_PAFGxP5CVWihdzuRjzX3XwobBgLGGW0UyBKSGowlBg/exec";

// Timer variables
unsigned long lastTime = 0;
unsigned long timerDelay = 10000; // Send data every 10 seconds

// DHT11 settings
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// MQ135 settings
#define MQ135PIN 34

// TEMT6000 settings
#define TEMTPIN 32

void setup() {
```

```

Serial.begin(115200);

// Initialize the DHT sensor dht.begin();

// Connect to Wi-Fi WiFi.begin(ssid,
password);
Serial.print("Connecting to Wi-Fi..."); while (WiFi.status()
!= WL_CONNECTED) {
    delay(1000); Serial.print(".");
}
Serial.println("\nConnected to Wi-Fi");
}

void loop() {
    if((millis() - lastTime) > timerDelay) {
        // Read DHT11 data

        float temperature = dht.readTemperature(); float humidity =
dht.readHumidity();

        // Read MQ135 data
        int gasValue = analogRead(MQ135PIN);

        // Read TEMT6000 data
        int lightValue = analogRead(TEMPIN);

        // Check if DHT readings failed
        if (isnan(temperature) || isnan(humidity)) { Serial.println("Failed to read from
DHT sensor!"); return;
        }

        // Log sensor readings to Serial Monitor Serial.println("Sensor Readings:");
        Serial.printf("Temperature: %.2f °C, Humidity: %.2f %%\n", temperature, humidity); Serial.printf("Gas: %d, Light:
%d\n", gasValue, lightValue);

        // Send data to Google Sheet if connected to Wi-Fi if (WiFi.status() ==
WL_CONNECTED) {
            HTTPClient http;

```

```

// Prepare the URL with sensor data
String url = String(serverName) +
             "?temperature=" + String(temperature) +
             "&humidity=" + String(humidity) +
             "&gas=" + String(gasValue) +
             "&light=" + String(lightValue);

// Send HTTP GET request
http.begin(url.c_str());
int httpResponseCode = http.GET();

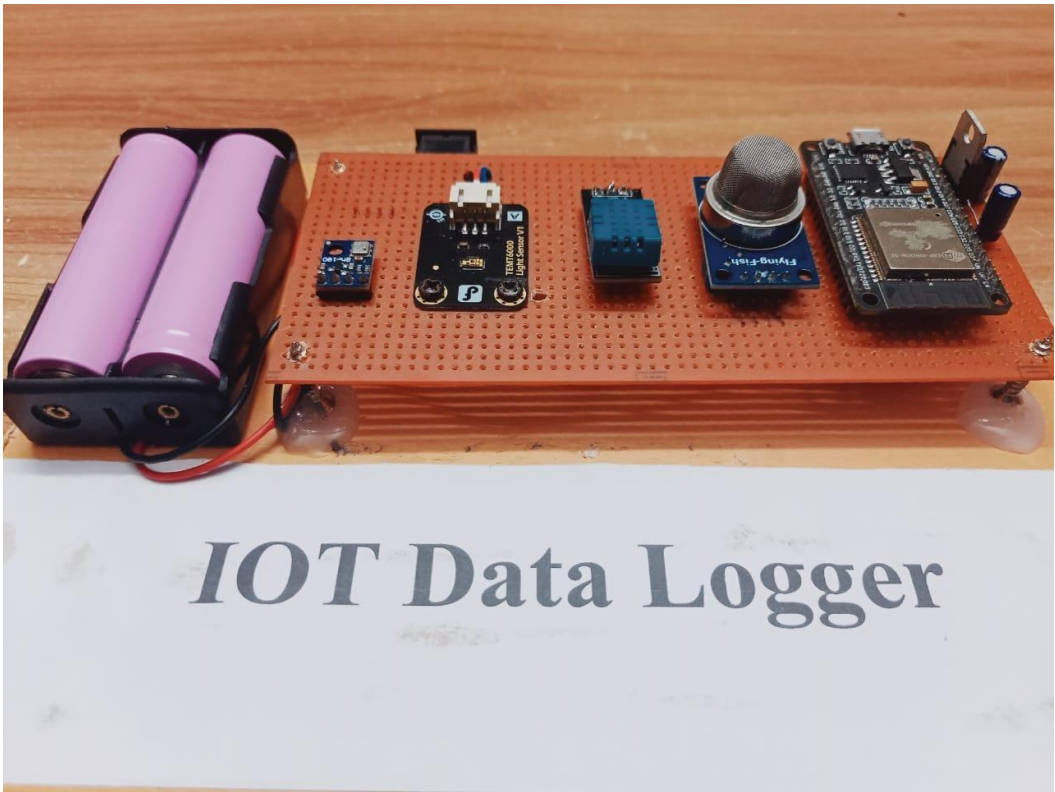
if (httpResponseCode > 0) {
    Serial.println("Data sent successfully!");
    String response = http.getString();
    Serial.println(response);
} else {
    Serial.print("Error sending data: ");
    Serial.println(httpResponseCode);
}

// End HTTP connection
http.end();
} else {
    Serial.println("Wi-Fi not connected");
}

// Update the timer
lastTime = millis();
}
}

```

Result:



IoT Data Logger

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Default...

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